

F=ma Extreme Re-entry Set

12 questions - 50 minutes - intentionally overcompressed

Difficulty target: approximately late-exam F=ma questions (20-25 range).

Use $g = 10 \text{ m/s}^2$ unless otherwise stated. No notes or formula sheet.

Instructions

- Work independently. Do not check answers during the set.
- Select one answer and one confidence level for every question.
- Mark any question that exceeds 4 minutes.
- The time limit is intentionally harsher than realistic F=ma pacing.
- Do not expect to finish every problem. Prioritize clean setups and intelligent skipping.

Student information

Name:

Date:

Start:

End:

Time remaining:

Questions exceeding 4 minutes

Setups I could not visualize

1. Projectile Rebounds from an Incline

A small ball is launched from a point on a fixed frictionless incline that makes an angle α with the horizontal. Its initial velocity is perpendicular to the incline. Each impact with the incline is perfectly elastic and frictionless. Let t_1 be the time from launch to the first impact, and t_2 the time between the first and second impacts. What is t_2/t_1 ?

A. 1

D. 4

B. 2

E. It depends on α

C. 3

Confidence:

C

U

G

Over 4 min:

Work / reasoning:

Final answer:

One-sentence reason / trigger:

2. Two-Stage Braking with a Delayed Brake

A car moves at speed v on a horizontal road. The driver reacts for a time τ before the brakes engage. Once engaged, the braking force has constant magnitude F . At speed v the total stopping distance is D . If the initial speed is $2v$ and the reaction time and braking force are unchanged, the total stopping distance is $10D/3$. What fraction of D is the original reaction distance $v\tau$?

A. $1/8$ D. $1/3$ B. $1/6$ E. $1/2$ C. $1/4$

Confidence:

C

U

G

Over 4 min:

Work / reasoning:

Final answer:

One-sentence reason / trigger:

3. Accelerating Wedge

A frictionless wedge of angle θ is accelerated horizontally to the left so that a small block remains at rest relative to the wedge. The incline rises to the right. What is the required acceleration magnitude?

A. $g \sin(\theta)$

D. $g \cot(\theta)$

B. $g \cos(\theta)$

E. $g / \sin(\theta)$

C. $g \tan(\theta)$

Confidence: C U G Over 4 min:

Work / reasoning:

Final answer:

One-sentence reason / trigger:

4. Coupled Movable-Pulley System

A movable pulley carries a mass $4m$. One end of its supporting rope is fixed to the ceiling; the other end passes over a fixed pulley and supports a mass m . A second independent rope passes under the movable pulley, with masses m and $3m$ hanging from its two ends. All ropes and pulleys are ideal. Immediately after release, what is the magnitude of the acceleration of the movable pulley?

A. $g/11$ D. $4g/11$ B. $2g/11$ E. $5g/11$ C. $3g/11$

Confidence: C U G Over 4 min:

Work / reasoning:

Final answer:

One-sentence reason / trigger:

5. Scale Reading for an Accelerating Atwood Machine

An ideal Atwood machine with masses $2m$ and $5m$ is suspended from a scale. The entire scale-pulley assembly is accelerating upward with acceleration a_0 relative to the ground. What does the scale read while the masses move?

A. $20m(g+a_0)/7$

D. $40m(g+a_0)/7$

B. $24m(g+a_0)/7$

E. $7m(g+a_0)$

C. $28m(g+a_0)/7$

Confidence: C U G Over 4 min:

Work / reasoning:

Final answer:

One-sentence reason / trigger:

6. Stacked Blocks with Force Applied to the Upper Block

A block of mass m rests on top of a block of mass $2m$. The floor is frictionless. The coefficient of static friction between the blocks is μ . A horizontal force F is applied to the upper block. What is the largest F for which the blocks move together?

A. $\mu m g$

D. $3\mu m g$

B. $3\mu m g/2$

E. $4\mu m g$

C. $2\mu m g$

Confidence:

C

U

G

Over 4 min:

Work / reasoning:

Final answer:

One-sentence reason / trigger:

7. Circular Motion with a Losing Contact Condition

A small block starts from rest at the top of a smooth sphere of radius R and slides on the outside surface. At what vertical drop h from the top does it lose contact with the sphere?

A. $R/6$ D. $R/2$ B. $R/4$ E. $2R/3$ C. $R/3$

Confidence: C U G Over 4 min:

Work / reasoning:

Final answer:

One-sentence reason / trigger:

8. Banked Curve with Friction at the Upper-Speed Limit

A car travels on a banked curve of radius R and banking angle θ . The coefficient of static friction is μ . At the maximum speed for no slipping, friction points down the slope. Which expression equals $v_{\text{max}}^2/(gR)$?

- A. $(\sin(\theta) - \mu \cos(\theta)) / (\cos(\theta) + \mu \sin(\theta))$ D. $\tan(\theta) + \mu$
B. $(\sin(\theta) + \mu \cos(\theta)) / (\cos(\theta) - \mu \sin(\theta))$ E. $\tan(\theta) / (1 - \mu)$
C. $(\cos(\theta) + \mu \sin(\theta)) / (\sin(\theta) - \mu \cos(\theta))$

Confidence: C U G Over 4 min:

Work / reasoning:

Final answer:

One-sentence reason / trigger:

9. Collision Followed by Circular Motion

A small block of mass m moving horizontally at speed v embeds in a block of mass $3m$ attached to a massless string of length L . The combined mass then swings upward in a vertical circle. What minimum v is required for the string to remain taut at the top?

A. $4\sqrt{gL}$

D. $8\sqrt{gL}$

B. $4\sqrt{2gL}$

E. $8\sqrt{5gL}$

C. $4\sqrt{5gL}$

Confidence: C U G Over 4 min:

Work / reasoning:

Final answer:

One-sentence reason / trigger:

10. Rod and Sliding End Constraint

A uniform rod of length L has its lower end constrained to slide on a frictionless horizontal floor and its upper end constrained to slide on a frictionless vertical wall. When the rod makes an angle θ with the floor, the lower end has speed v . What is the speed of the rod's center of mass?

A. $v/2$

D. $v/(2 \sin(\theta) \cos(\theta))$

B. $v/(2 \sin(\theta))$

E. $v/(2 \cos^2(\theta))$

C. $v/(2 \cos(\theta))$

Confidence: C U G Over 4 min:

Work / reasoning:

Final answer:

One-sentence reason / trigger:

11. Two-Dimensional Pursuit Acceleration

A target moves east at constant speed u . A pursuer moves at constant speed v and always points directly toward the target. At an instant when the separation vector points due north and has magnitude L , what is the magnitude of the pursuer's acceleration?

A. u^2/L

D. $(v-u)^2/L$

B. uv/L

E. $uv/(2L)$

C. v^2/L

Confidence:

C

U

G

Over 4 min:

Work / reasoning:

Final answer:

One-sentence reason / trigger:

12. Gravitational Tunnel Oscillation

A narrow straight tunnel is drilled through the center of a uniform spherical planet of radius R and surface gravity g . A small object is released from rest at the surface into the tunnel. Neglect rotation and drag. How long does it take to first reach the center?

A. $(\pi/2) \sqrt{R/g}$

D. $(\pi/2) \sqrt{2R/g}$

B. $\pi \sqrt{R/g}$

E. $2\pi \sqrt{R/g}$

C. $\pi \sqrt{R/(2g)}$

Confidence: C U G Over 4 min:

Work / reasoning:

Final answer:

One-sentence reason / trigger:

Final Answer Summary

Record answers only after completing the set.

1.

Conf.

>4m
7.

Conf.

>4m
2.

Conf.

>4m
8.

Conf.

>4m
3.

Conf.

>4m
9.

Conf.

>4m
4.

Conf.

>4m
10.

Conf.

>4m
5.

Conf.

>4m
11.

Conf.

>4m
6.

Conf.

>4m
12.

Conf.

>4m

Hardest problem:

Most uncertain topic:

What caused lost time?

Concepts to review